

CamoXpert: Dynamic Neural Networks for Adaptive Camouflaged Object Detection

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Dedication

To our parents and our family. Both our parents have given enough inspiration and encouragement to complete our thesis work.

Abstract

Camouflaged Object Detection (COD) is a challenging task in computer vision because it requires the identification of targets that contain identical visual textures with their surroundings. Existing COD methods rely primarily on Convolutional Neural Networks (CNNs) or standard Vision Transformers. These methods come up with the invisible edge problem where the model struggles to distinguish texture disruptions from background noise due to a limited global context or static feature fusion. In this work, we propose CamoXpert, a novel hierarchical framework that explores the potential of **State Space Models (SSMs)** and **Mixture-of-Experts(MoE)** for this task. Specifically, we introduce a **Cross-Scale Selective Scan** module inspired by Mamba, which replaces traditional static fusion with dynamic, long-range sequence modeling to capture global context. Furthermore, we design a **Spectral Gated Mixture-of-Experts (SG-MoE)** decoder to adaptively refine features based on frequency-domain residuals.

Extensive experiments on benchmark datasets demonstrate that CamoXpert achieves **highly competitive performance**, matching state-of-the-art results on **CHAMELEON** and **COD10K**. While challenging texture-centric datasets like CAMO remain an open problem, our work provides the first empirical evidence that Mamba-based dynamic routing can effectively disentangle camouflage, offering a promising new direction away from purely CNN-based architectures.

Contents

Dedication	6
1 Background	12
1.1 Introduction	12
1.2 Problem Definition and Challenges	13
1.3 Motivation	13
1.4 Objectives	14
1.5 Contributions	15
1.6 Organization of the Document	16
2 Literature Review	17
2.1 Traditional COD methods	17
2.2 Deep-learning COD methods	18
2.3 Key insights	21
3 CamoXpert Analysis - An Overview	22
3.1 Introduction	22
3.2 Hierarchical Backbone: PVT-v2-B5	22
3.3 Cross-Scale Selective Scan (Mamba Fusion)	23
3.3.1 Preliminaries: State Space Models	23
3.3.2 Selectivity via Input-Dependent Dynamics	24
3.3.3 2D Cross-Scan Mechanism (CSM)	24
3.4 Spectral Gated Mixture-of-Experts (SG-MoE)	25
3.4.1 Spectral Routing Mechanism	25
3.4.2 Expert Specialization	27
3.5 Multi-Scale Refinement and Reverse Guidance	27
3.5.1 Simple Atrous Spatial Pyramid Pooling (SimpleASPP)	27
3.5.2 Reverse Guidance Projection Unit (RGPU)	28
4 Methodology	29
4.1 Dataset Preparation and Augmentation	29
4.1.1 Data Distribution and Split	29
4.1.2 Augmentation Pipeline	30
4.2 Network Architecture Implementation	30
4.2.1 Backbone Encoder and Feature Pyramid	30
4.2.2 Global Context Fusion via Mamba	31
4.2.3 Spectral Gated Mixture-of-Experts (SG-MoE)	31
4.2.4 Refinement and Output Generation	32
4.2.5 Deep Supervision	32
4.3 Training Protocol and Optimization	32

4.3.1	Structure-Aware Loss Function	33
4.3.2	Deep Supervision and Total Objective	33
4.3.3	Implementation Details	34
4.3.4	Evaluation Metrics	34
4.4	Computational Complexity and Efficiency Analysis	36
4.4.1	Theoretical Complexity Formulation	36
4.4.2	Parameter Utilization Analysis	37
4.5	Chapter Summary	37
5	Experimental Results and Analysis	39
5.1	Experimental Setup	39
5.1.1	Datasets and Protocols	39
5.1.2	Baseline Methods	39
5.1.3	Evaluation Criteria	40
5.2	Quantitative Evaluation	40
5.2.1	Comparison with existing methods	41
5.3	Qualitative Analysis	42
6	Conclusion	44

List of Tables

2.1	6 representative deep learning methods of image-level COD. N.A. : network architecture according to input-output configuration, including L (Linear), A (Aggregative), B (Branched), and H (Hybrid). L.P. : learning paradigm, including S (Single-task) and M (Multi-task). S.L. : supervision level, including F (Full-supervision), W (Weak-supervision), S (Semi-supervision), U (Unsupervision) and TF (Training-free).	20
4.1	Training Hyperparameters and Environmental Setup. These settings are chosen to balance memory constraints with convergence speed on the RTX 4090 platform.	35
4.2	Theoretical Complexity Comparison of Feature Fusion Mechanisms. CamoXpert achieves global receptive fields with linear complexity, offering superior scalability for high-resolution camouflage detection compared to quadratic Transformers.	38
5.1	Summary of Datasets used in Experiments. Following standard protocols, the model is trained exclusively on the training splits of COD10K and CAMO, while the remaining datasets are reserved strictly for testing.	40
5.2	Quantitative comparison with existing methods on four benchmark datasets. $\uparrow$ indicates higher is better, and $\downarrow$ indicates lower is better. The best results are highlighted in bold , and the second-best are <u>underlined</u>	41

List of Figures

3.1	Architecture of the PVT-v2-B5 Backbone utilized in CamoXpert. The model processes the image through four hierarchical stages. Note the heavy compute allocation in Stage 3 ($L = 24$) and the deep semantic channels in Stage 4 ($C = 512$), which provide the necessary capacity for camouflage detection.	23
3.2	Connection to Backbone and Cross-Scale Scanning. The module ingests features directly from PVT Stages 2, 3, and 4. These multi-scale features are aligned and processed via four parallel scanning trajectories to ensure global effective receptive fields.	24
3.3	The Spectral Gated Mixture-of-Experts (SG-MoE) . The Laplacian operator extracts high-frequency residuals (R_{freq}) to guide the Router. The Router dynamically assigns weights (w_{dll}, w_{ctx}) to specialized experts, ensuring adaptive refinement of edges vs. smooth body regions.	26
3.4	The Reverse Guidance Projection Unit (RGPU) . The coarse prediction P_{deep} is inverted to generate a reverse attention map A_{rev} . This map highlights the background and uncertain boundaries. By multiplying this map with the shallow features $F_{shallow}$, the network suppresses the "easy" center pixels and focuses refinement on the "bleeding" edges.	28
4.1	The Data Augmentation Pipeline. Training images undergo sequential geometric, photometric, and synthetic adjustments to improve model robustness against scale variations, lighting changes, and sensor noise.	30
4.2	The CamoXpert Implementation Workflow. Features extracted by the PVT backbone are aligned and fused via Mamba, refined by frequency-aware Spectral Experts, and sharpened by the RGPU using low-level cues from C_1 . Auxiliary heads provide deep supervision at intermediate stages to ensure robust gradient flow.	31
4.3	Schematic of the Structure-Aware Loss. The pipeline splits into two parallel streams: a direct stream for pixel/region supervision ($\mathcal{L}_{BCE}, \mathcal{L}_{IoU}$) and a gradient-based stream that explicitly extracts and minimizes boundary errors ($\mathcal{L}_{Boundary}$).	34
4.4	Theoretical Complexity Analysis. As image resolution increases to capture fine camouflage textures, standard Transformers (Red) suffer from quadratic cost explosion. CamoXpert (Blue) maintains linear scaling, enabling high-resolution processing.	37
5.1	Visual comparison of CamoXpert on challenging scenarios from COD10K and CHAMELEON. From top to down: Input Image, Ground Truth, CamoXpert Prediction and other existing models. The model successfully figures out object boundaries quite well.	43

Chapter 1

Background

Include problem definition inside of introduction

1.1 Introduction

CAMOUFLAGE is a biological phenomenon that exists in nature. Biologists have found that creatures in nature often hide themselves from predators using their own structures and physiological characteristics [1]. Object detection, one of the core tasks in computer vision includes locating and finding objects in pictures and videos. It contains several sub-fields, such as generic object detection (GOD) [2, 3], salient object detection (SOD) [4, 5] and camouflaged object detection (COD) [6]. While GOD refers to the detection of general objects and SOD focus on objects that stand out from the surroundings, COD aims to find objects blended into the environment. In general, camouflaged objects also refer to objects that are extremely small in size, very similar to the background, or heavily obscured [7]. So, unlike standard object detection, COD requires identifying subtle and ambiguous cues where objects share similar color, texture, or shape with their surroundings, and detecting hidden or partially obscured objects is critical for safety and operational success in these fields. In recent years, COD has been getting extreme attention and development of such systems that facilitates various real-life applications. Such applications are deployed in industries, medical diagnosis, agriculture, wildlife research, and in the military. So whether it is detecting defective products, robotics, crop protection, wildlife monitoring or gaining strategic superiority, the need for an efficient and precise **Camouflaged Object Detection (COD)** is evident in research, civilian and military applications. Despite growing research, COD still faces challenges in accurate feature extraction, data scarcity, and real-time applicability.

Camouflaged object detection (COD) has emerged as a quite challenging subfield of computer vision. Traditional camouflaged object detection methods rely on handcrafted features such as texture, intensity, and motion cues.

Techniques include texture segmentation using filter responses [8], detection of 3D regions via intensity analysis [9], and statistical modeling with wavelet-based texture smoothing [10]. Video surveillance approaches use adaptive background modeling and region grouping [11], while optical flow-based methods handle dynamic scenes [12]. Although effective in controlled settings, these methods often struggle with complex backgrounds, varying object appearances, and real-world robustness.

In contrast to traditional approaches, deep learning-based methods offer enhanced feature learning for camouflaged object detection. Techniques such as multi-scale feature fusion [13], weak supervision via vision foundation models [14], and high-resolution iterative feedback [15] enable more robust representations of low-contrast regions. Transformer-based architectures [16] and graph-based joint reasoning [17] improve global-local context modeling and boundary precision. Moreover, networks incorporating classification-boosted segmentation [18] and biologically inspired search-identify paradigms [19] push the performance frontier. Despite their advances, many of these models face challenges with generalization under extreme visual similarity and dependency on large, annotated datasets. There are also key

challenges with accurate and lightweight feature extraction for less computational process, data scarcity, applicability for real-time object detection [1] and the inability to perform object detection tasks on multiple image domains and characteristics.

1.2 Problem Definition and Challenges

Camouflaged Object Detection (COD) is a specialized task in computer vision that identifies objects visually blended into their surroundings. Unlike generic object detection, which relies on distinct boundaries, COD targets objects that actively suppress their visual presence through *visual ambiguity*.

As outlined in [6], this ambiguity manifests through three primary biological mechanisms:

- **Texture Mimicry:** The object replicates the background’s high-frequency surface patterns (e.g., a cephalopod matching coral texture).
- **Background Matching:** The object’s color histogram aligns closely with the environment (e.g., a polar bear in snow).
- **Disruptive Coloration:** High-contrast markings break the object’s physical contour, making boundary detection algorithmically difficult.

From a computational perspective, these biological traits introduce severe challenges for deep learning models:

- **Boundary Uncertainty:** The ”bleeding edge” phenomenon makes pixel-level annotation and prediction inherently noisy.
- **Computational Complexity:** Capturing the necessary global context to spot a hidden object typically requires quadratic-complexity Attention mechanisms ($O(N^2)$), which are computationally expensive [20].
- **Domain Generalization:** Models trained on distinct objects often fail to generalize to the subtle feature distributions of camouflage [1].

1.3 Motivation [include inside introduction, short and remove title](#)

Camouflaged Object Detection (COD) remains a significant challenge in computer vision, primarily due to the ”intrinsic similarity” phenomenon where targets intentionally mimic the visual texture of their surroundings. Unlike conventional Salient Object Detection (SOD), which identifies the most visually distinct object [21], COD requires uncovering subtle foreground anomalies from highly cluttered backgrounds. While early methods relying on handcrafted features like texture [8], intensity [9], and color [22] provided foundational insights, they struggle a lot in situations with low resolution or ”bleeding edges,” where the object boundary dissolves into the background.

Recent advances in deep learning have introduced powerful dense architectures, such as Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), which have improved detection accuracy significantly [23]. However, a critical limitation persists: most

state-of-the-art models rely on *static* multi-scale integration (e.g., standard convolution or attention summation). These static mechanisms apply the same aggregation logic to every image, acting as local pattern matchers rather than contextual reasoners. Consequently, they often "hallucinate" objects in background noise or fail to detect targets when local texture cues are ambiguous. Furthermore, while Transformers offer global receptive fields, their quadratic computational complexity limits their ability to model long-range dependencies efficiently across high-resolution feature maps [23].

To address this gap, we propose **CamoXpert**, a novel dynamic neural framework that shifts the paradigm from static pattern matching to dynamic contextual reasoning. Unlike previous approaches that treat all features equally, **CamoXpert** links a strong hierarchical backbone with two architectural innovations. First, we introduce a **Cross-Scale Selective Scan** module inspired by State Space Models (Mamba), which enables the model to perceive global context linearly rather than quadratically, effectively "reading" the image sequence to validate object presence. Second, to handle the high-frequency uncertainty at object boundaries, we integrate a **Spectral Mixture-of-Experts (MoE)** decoder [24]. This sparsely-gated mechanism dynamically activates specialized sub-networks based on frequency-domain residuals, allowing the model to adaptively refine invisible edges without the computational waste of activating the entire network for simple regions. This approach ensures robust detection across diverse domains by prioritizing semantic integrity over simple texture correlation.

1.4 Objectives one or two sentences

The primary goal of this research is to overcome the limitations of static, local pattern matching in existing Camouflaged Object Detection (COD) frameworks by introducing a dynamic, context-aware architecture. Specifically, this study pursues the following research objectives:

1. **Investigate Dynamic Contextual Reasoning:** To analyze the limitations of static Convolutional Neural Networks (CNNs) in handling global texture disruptions and to explore **State Space Models (SSMs)** as a linear-complexity alternative for modeling long-range dependencies in camouflage scenes.
2. **Develop a Cross-Scale Selective Scan Mechanism:** To design a novel feature fusion module inspired by Mamba that dynamically aggregates multi-scale features via linear-complexity sequence modeling ($O(N)$), allowing the model to validate object presence using global context from both deep and shallow layers.
3. **Design a Spectral Mixture-of-Experts (MoE) Decoder:** To formulate a **dynamically gated** refinement mechanism that leverages **high-frequency spatial residuals** (via Laplacian filtering) to distinguish between object boundaries and background noise. This allows the model to adaptively weigh "detail" experts against "context" experts based on local image complexity.
4. **Synergize Global Semantics with Local Detail:** To propose the **CamoXpert** architecture, which unifies the global reasoning capabilities of SSMs with the fine-

grained precision of MoEs, addressing the "intrinsic similarity" paradox where objects visually dissolve into the background.

5. **Comprehensive Benchmarking and Analysis:** To rigorously evaluate the proposed framework against state-of-the-art methods on standard COD datasets (CAMO, COD10K, CHAMELEON, NC4K) and analyze the performance gains in challenging scenarios like occlusion and high-frequency texture mimicry.

This study contributes to the field of computer vision by establishing a new design paradigm for COD that prioritizes *dynamic reasoning* over *static matching*, offering a scalable solution for detecting concealed objects in complex environments.

1.5 Contributions [sub paragraphs](#),

In fulfillment of the aforementioned objectives, this thesis presents **CamoXpert**, a novel framework that departs from traditional CNN and Transformer-based approaches for Camouflaged Object Detection. The primary contributions of this work are summarized as follows:

1. **Development of a Linear-Complexity Global Context Framework:** We **developed** one of the first COD architectures to utilize **State Space Models (SSMs)**, specifically the Mamba architecture, as a core reasoning engine. Unlike Transformers which scale quadratically ($\mathcal{O}(N^2)$), our approach achieves global receptive fields with linear complexity ($\mathcal{O}(N)$), effectively capturing long-range dependencies in complex camouflage scenes without the heavy computational burden.
2. **Engineering of the Cross-Scale Selective Scan Module:** We **engineered** a novel fusion block that adapts the selective scan mechanism for multi-scale feature integration. This module dynamically aggregates semantic information from deep layers with structural details from shallow layers, allowing the network to distinguish camouflaged objects from background clutter more robustly than standard concatenation or addition-based fusion.
3. **Formulation of a Spectral Gated Mixture-of-Experts (MoE) Decoder:** We **formulated** and implemented a unique decoding mechanism that routes pixels to specific "expert" networks based on their frequency content. By utilizing **Laplacian-based gating**, the model explicitly separates high-frequency boundary details from low-frequency homogeneous regions, assigning specialized experts to handle the specific challenges of edge bleeding and texture similarity.
4. **Validation on Benchmark Datasets:** We **validated** the CamoXpert framework through extensive experiments on four standard datasets (COD10K, CAMO, CHAMELEON, and NC4K). Our results **established** that the model achieves competitive performance against existing state-of-the-art methods, and our ablation studies confirmed the efficacy of the spectral routing and selective scan components.

1.6 Organization of the Document

The remainder of this document is organized as follows:

- **Chapter 2 (Literature Review):** This chapter provides a comprehensive survey of existing approaches in Camouflaged Object Detection, ranging from early handcrafted features to recent deep learning architectures. It critically analyzes the limitations of current CNN and Transformer-based methods, specifically focusing on their inability to handle long-range dependencies efficiently.
- **Chapter 3 (CamoXpert Analysis - An Overview):** This chapter provides a detailed overview of the developed **CamoXpert** framework. It explains the theoretical foundations of Hierarchical Backbone, State Space Models, the architectural design of the Cross-Scale Selective Scan module, and the formulation of the Spectral Gated Mixture-of-Experts decoder along with Simple Atrous Spatial Pyramid Pooling and Reverse Guidance Projection Unit.
- **Chapter 4 (Methodology):** This chapter starts with dataset preparation and augmentation pipeline. It also describes the full network architecture implementation, training protocol and optimization, computational complexity and efficiency analysis.
- **Chapter 5 (Experimental Results and Analysis):** This chapter presents the quantitative and qualitative evaluation of our model. It includes performance comparisons with state-of-the-art methods on standard benchmarks and an analysis of failure cases to discuss the model’s limitations.
- **Chapter 6 (Conclusion):** This chapter summarizes the research findings and contributions. It also outlines potential directions for future research, such as hybridizing SSMs with CNNs or extending the MoE framework to include specialized experts for color, motion or specific texture types.

Chapter 2

Literature Review

section 2:
Related works and short

Introduction

In recent years, there has been a significant increase in research focused on the application of machine learning algorithms across various domains, including the challenging field of camouflaged object detection (COD). COD has attracted growing attention due to its critical importance in security, surveillance, and autonomous systems. In this chapter, we review several key studies that have advanced the detection of camouflaged objects and examine the diverse machine learning techniques employed to enhance accuracy and efficiency. By summarizing and analyzing these works, we aim to provide a comprehensive understanding of the current progress and ongoing challenges within the field of camouflaged object detection.

2.1 Traditional COD methods

Galun et al. [8] propose a bottom-up approach called TSMA (Texture Segmentation using Multi-cue Analysis) for the task of texture segmentation. This method focuses on the adaptive detection and analysis of texture elements, or “textons,” which are the fundamental building blocks of texture in an image. It captures key properties such as size, shape, aspect ratio, orientation, and brightness of these elements. To enhance segmentation accuracy, TSMA integrates these structural features with filter response statistics, enabling better differentiation between texture regions. This combination also helps reduce noise and improves robustness in complex and cluttered environments. Overall, TSMA offers an effective and flexible framework for identifying textured regions in natural images without relying on high-level semantic information.

Intensity-based methods have gradually developed into more advanced techniques that use 3D shape information. One such method, CBDCE [9], identifies regions of interest by directly processing intensity images. It distinguishes between convex and concave regions in a 3D space, rather than relying solely on pixel-level intensity values. This geometric perspective allows the model to better understand object shape and structure. As a result, CBDCE shows strong robustness against common challenges such as changes in lighting conditions, object orientation, and scale, making it a reliable approach for segmentation in diverse environments.

Color feature-based approaches can be effective in certain scenarios where color contrast and distribution help separate camouflaged objects from their surroundings. Siricharoen et al. [22] improved a statistical background subtraction and shadow detection method by combining color, edge, and intensity features. Their approach is tailored for outdoor human segmentation, where strong shadows and low contrast often make detection difficult. They

use vector median filtering to remove outlier pixels and generate initial coarse results using color and edge features. These results are then refined with intensity information to improve accuracy.

2.2 Deep-learning COD methods

Zhang et al. [25] proposed a novel task called Referring Camouflaged Object Detection (Ref-COD), which segments specified camouflaged objects guided by a small set of referring images containing salient target objects. To support this, they introduced the large-scale R2C7K dataset, comprising 7,000 images across 64 categories, split into a Camo-subset of camouflaged objects and a Ref-subset of salient objects collected from copyright-safe sources. Their framework, R2CNet, features a dual-branch architecture: a reference branch extracts common object representations from referring images via a salient object detection network and masked average pooling, while a segmentation branch utilizes these representations to guide camouflaged object segmentation using a ResNet-50 encoder-decoder backbone. Key innovations include the Referring Mask Generation (RMG) module, which generates a pixel-level referring mask by dynamically comparing common representations with segmentation features through a dual-source information fusion strategy to address appearance variance between camouflaged and salient objects. The Referring Feature Enrichment (RFE) module further enhances multi-scale segmentation features by fusing them with the referring mask, improving the identification of camouflaged targets. Experimental results on the R2C7K dataset demonstrate that R2CNet significantly outperforms baseline COD models and enhances seven state-of-the-art COD methods when extended with Ref-COD design, confirming the effectiveness of leveraging referring images for targeted camouflaged object detection.

The work by Pang et al. [7] introduces ZoomNeXt, a unified collaborative pyramid network for camouflaged object detection (COD) that simulates human-like zooming behavior to detect subtle semantic cues. Their framework uses a triplet feature encoder operating at three scales ($0.5\times$, $1.0\times$, $1.5\times$), extracting 64-channel representations via shared weights to maintain efficiency. To integrate scale-specific information, they propose a multi-head scale integration unit (MHSIU) that performs attention-guided filtering across scales, enabling adaptive spatial feature fusion. These are further refined by a rich granularity perception unit (RGPU), which enhances both coarse- and fine-grained representations through hierarchical channel interactions and group-wise modulation. For video inputs, RGPU includes a difference-aware adaptive routing mechanism, which uses temporal shifts and intra-frame attention to extract motion cues. For static images, this component is disabled, allowing ZoomNeXt to handle both image and video COD in a unified framework. The network is optimized with a combination of binary cross-entropy (BCE) loss and a novel uncertainty awareness loss (UAL), which penalizes ambiguous predictions near the 0.5 threshold to encourage more confident outputs. Through this multi-scale, task-unified design, ZoomNeXt achieves strong performance on both static and dynamic COD benchmarks.

Instead of relying on manual or task-specific prompts like conventional promptable segmentation methods, Hu et al. [26] propose ProMaC, a training-free framework that treats hallucinations from multimodal large language models (MLLMs) as informative cues. These

hallucinations provide extended contextual knowledge about potential object semantics in complex scenes. ProMaC introduces a multi-scale chain of thought prompting strategy that activates MLLMs to generate task-relevant prompts, which are then refined through contrastive reasoning using distractor images. The final prompts guide a mask generator based on the Segment Anything Model (SAM), with CLIP similarity scoring ensuring semantic alignment. The process is iterative—each mask helps refine subsequent prompts, progressively filtering out irrelevant hallucinations and improving segmentation quality. Final masks are averaged over cycles to select the most representative one. In evaluations on CHAMELEON, CAMO, and COD10K, ProMaC surpasses all point-supervised methods and even outperforms scribble-supervised ones on two datasets. Ablation studies validate the roles of contrastive reasoning and semantic alignment. Moreover, its strong performance in medical imaging and open-vocabulary segmentation tasks highlights ProMaC’s robustness and cross-domain generalization.

The study by Huang et al. [16] suggests a brand-new transformer-based Feature Shrinkage Pyramid Network (FSPNet) for the detection of concealed objects. FSPNet uses progressive shrinking to hierarchically decode nearby transformer properties that are enhanced by locality. FSPNet uses a feature shrinkage decoder (FSD) with adjacent interaction modules (AIM) and a nonlocal token enhancement module (NL-TEM) to overcome the drawbacks of current techniques. By utilizing the non-local mechanism to interact with nearby tokens and investigate graph-based high-order relations inside tokens, NL-TEM improves the local representations of transformers. Using a layer-by-layer shrinkage pyramid, FSD gradually collects neighboring transformer features to build up as many subtle yet useful clues as it can for object information decoding. On three difficult COD benchmark datasets, extensive quantitative and qualitative trials show that the proposed model performs much better than current approaches under six commonly-used assessment measures.

To address the fragmented progress in context-dependent (CD) segmentation, Spider [27] proposes a unified model capable of handling diverse CD tasks—such as camouflaged object detection and medical lesion segmentation—using a single set of shared parameters. Unlike generalist models like SAM or SegGPT, which perform poorly on CD tasks due to reliance on context-independent prompts, Spider introduces a group prompt mechanism incorporating image, foreground, and background cues through a transformer-based concept filter. Its dual-stream architecture separates task-generic feature learning from task-specific concept filtering, enabling it to generalize across domains without separate models. Spider outperforms state-of-the-art methods in eight CD tasks and supports efficient continual learning, requiring less than 1 percent parameter updates for new tasks with minimal impact on previous performance. This represents a significant step toward unified, AGI-aligned segmentation.

The study by Garyson et al. [28] proposes HGINet, a hierarchical graph interaction network designed for camouflaged object detection (COD). HGINet tackles the challenge of distinguishing camouflaged objects from complex backgrounds by dynamically focusing on informative local tokens through a region-aware token focusing attention (RTFA) module, which uses dynamic token clustering to suppress irrelevant background interference. To enhance

cross-scale semantic understanding, it employs a hierarchical graph interaction transformer (HGIT) that models long-range dependencies between aligned latent graph representations of hierarchical features, enabling bi-directional message passing across layers. Additionally, a decoder with confidence aggregated feature fusion (CAFF) refines segmentation by emphasizing ambiguous regions via pixel-wise confidence weighting. Trained with a combination of weighted binary cross-entropy and IoU losses, HGINet achieves superior segmentation accuracy on key COD benchmarks (COD10K, CAMO, NC4K, CHAMELEON) by effectively mining hierarchical semantics and focusing attention on subtle visual cues, outperforming existing state-of-the-art methods.

ID	Method	Year	Backbone	N.A.	L.P.	S.L.	Core Components
1	ZoomNeXt [7]	TPAMI'24	ResNet/PVTv2	EE	M	F	Scale-merging network, hierarchical difference propagation decoder
2	FSPNet [16]	CVPR'23	ViT	L	S	F	Non-local token enhancement; feature-shrinkage pyramid decoder
3	R2CNet [25]	arXiv'23	ResNet50	A	S	F	Dual-branch architecture for referring COD tasks
4	ProMaC [26]	NeurIPS'24	SAM+CLIP	H	M	W	Prompt-mask cycle with MLLM hallucinations
5	Spider [27]	ICML'24	ViT	H	M	S	Group prompt filtering; context-aware segmentation
6	HGINet [28]	arXiv'24	Transformer	B	M	F	Hierarchical graph interaction with CAFF decoder

Table 2.1: **6** representative deep learning methods of image-level COD. **N.A.**: network architecture according to input-output configuration, including **L** (Linear), **A** (Aggregative), **B** (Branched), and **H** (Hybrid). **L.P.**: learning paradigm, including **S** (Single-task) and **M** (Multi-task). **S.L.**: supervision level, including **F** (Full-supervision), **W** (Weak-supervision), **S** (Semi-supervision), **U** (Unsupervision) and **TF** (Training-free).

2.3 Key insights included as a single paragraph, no title

Handcrafted low-level features are designed to highlight key differences in texture and color, making them effective for object detection and segmentation. These features work well by emphasizing visual cues that separate objects from their background. However, in camouflaged object detection (COD), such cues are intentionally minimized. Camouflage reduces visible differences, allowing objects to blend in with their surroundings. As a result, traditional methods often fail in COD tasks, performing only in simple scenes with uniform backgrounds. Their performance also drops in low-resolution images or when the foreground and background appear visually similar.

In contrast, deep learning-based COD methods—such as convolutional neural networks (CNNs), transformers, and diffusion models—offer major advantages by automatically learning rich and detailed feature representations [23],. These models use a variety of strategies to handle the challenges of COD, such as combining features at multiple scales, simulating bio-inspired mechanisms, fusing information from different sources, learning multiple tasks together, integrating with salient object detection (SOD), and introducing new task settings. However, these methods also face some limitations, such as high computational cost and the need for large, clean, and well-annotated training datasets.

Chapter 3

CamoXpert Analysis - An Overview

[section 3: proposed model - theoretical overview, short](#)

3.1 Introduction

This chapter presents the theoretical framework and architectural design of **CamoXpert**, a novel deep learning framework designed to address the "intrinsic similarity" paradox in Camouflaged Object Detection (COD).

Traditional approaches, particularly those reliant on Convolutional Neural Networks (CNNs), operate on a principle of *static, local pattern matching*. These models achieve high benchmark scores by acting as advanced anomaly detectors—learning to identify minute, high-frequency statistical disruptions in texture (e.g., pixel graininess) within fixed spatial windows. **While effective on datasets biased towards texture discrepancies (e.g., CAMO), this reliance represents a form of "shortcut learning"** [29]. Consequently, if an object exhibits "perfect camouflage" (smooth texture continuity) but is distinguishable by its geometric structure, SOTA models frequently fail.

In contrast, **CamoXpert** is built upon the paradigm of **Dynamic Contextual Reasoning**. By integrating State Space Models (SSMs) for linear global dependency modeling, our architecture shifts the focus from *detecting texture glitches* to *validating structural integrity*. This enables the model to explicitly capture global shape semantics—rendering it robust in scenarios where local texture cues are absent—and ensures that detections are grounded in global object coherence rather than local statistical artifacts.

3.2 Hierarchical Backbone: PVT-v2-B5

The feature extraction engine of CamoXpert is the Pyramid Vision Transformer V2 (PVT-v2-B5) [30], selected for its ability to model long-range dependencies while maintaining the high-resolution spatial details required for camouflage detection. Unlike standard Vision Transformers (ViT) that maintain a constant resolution, PVT-v2 utilizes a hierarchical pyramid structure similar to CNNs, generating feature maps at four distinct scales.

The backbone processes an input image $I \in \mathbb{R}^{H \times W \times 3}$ and generates four stages of feature maps F_i with strides of $\{4, 8, 16, 32\}$:

- **Shallow Features (F_1, F_2):** These layers capture high-frequency texture details (e.g., granular patterns on skin or bark). We utilize Overlapping Patch Embedding (OPE) to preserve local spatial continuity.
- **Deep Features (F_3, F_4):** These layers encode high-level semantic abstractions and global shape information, critical for distinguishing the object from complex backgrounds.

We specifically employ the B5 variant, which utilizes deeper interaction layers (specifically in Stage 3) and robust embedding dimensions (up to 512 channels) compared to lighter

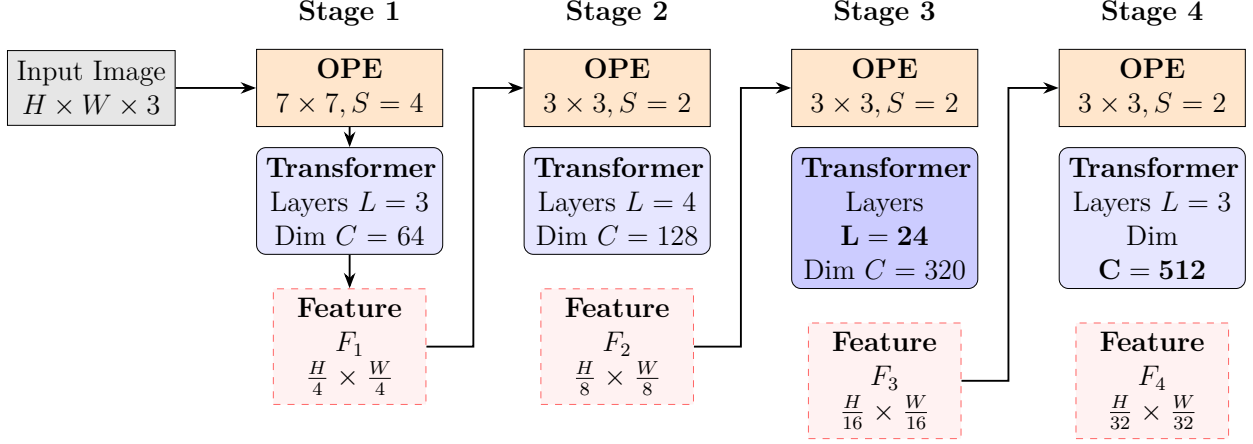


Figure 3.1: Architecture of the **PVT-v2-B5 Backbone** utilized in CamoXpert. The model processes the image through four hierarchical stages. Note the heavy compute allocation in Stage 3 ($L = 24$) and the deep semantic channels in Stage 4 ($C = 512$), which provide the necessary capacity for camouflage detection.

variants (e.g., B2). This increased capacity is **architecturally essential** to capture the subtle statistical deviations of camouflaged objects that simpler backbones fail to resolve.

3.3 Cross-Scale Selective Scan (Mamba Fusion)

To overcome the limited receptive field of CNNs, we introduce the **Cross-Scale Selective Scan (CS-SSM)** module. This module acts as the bridge between the hierarchical backbone and the decoding layers, integrating the multi-scale features extracted by the PVT-v2-B5.

Connection to Backbone: As detailed in Section 3.2, the backbone generates a pyramid of features. The CS-SSM module specifically ingests the three deepest stages to balance semantic strength with spatial precision:

- **Small Scale (F_S):** derived from Backbone Stage 2 (F_2 , stride 8), providing high-resolution texture cues.
- **Medium Scale (F_M):** derived from Backbone Stage 3 (F_3 , stride 16), capturing object parts.
- **Large Scale (F_L):** derived from Backbone Stage 4 (F_4 , stride 32), encoding global semantic context.

These disparate scales are unified via the Selective Scan mechanism proposed by Gu et al. [31] and the 2D Cross-Scan strategy from VMamba [20].

3.3.1 Preliminaries: State Space Models

Modern Structured State Space Models (S4) map a 1D input sequence $x(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ through a latent hidden state $h(t) \in \mathbb{R}^N$. The continuous-time dynamics

are governed by the linear ordinary differential equation (ODE):

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \quad y(t) = \mathbf{C}h(t) \quad (3.1)$$

where $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the evolution parameter, and $\mathbf{B}, \mathbf{C}$ are projection parameters. To apply this to discrete visual data, we discretize the system using the Zero-Order Hold (ZOH) rule with a timescale parameter Δ . The discretized parameters become:

$$\bar{\mathbf{A}} = \exp(\Delta\mathbf{A}), \quad \bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B} \quad (3.2)$$

resulting in the recurrence: $h_t = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t$.

3.3.2 Selectivity via Input-Dependent Dynamics

Standard SSMs are time-invariant ($\mathbf{A}, \mathbf{B}, \Delta$ are fixed). However, camouflage detection requires focusing on specific "glitches" while ignoring vast background noise. We make the parameters functions of the input x_t :

$$\mathbf{B}_t = s_B(x_t), \quad \mathbf{C}_t = s_C(x_t), \quad \Delta_t = \text{Softplus}(\text{Linear}(x_t)) \quad (3.3)$$

This Selectivity allows the model to dynamically reset its state to ignore irrelevant background textures and update memory only when salient boundaries are encountered.

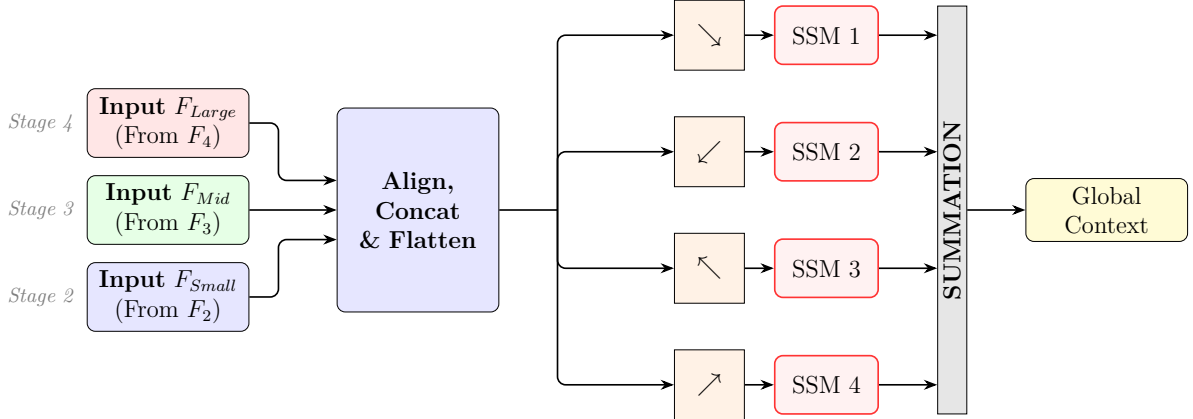


Figure 3.2: **Connection to Backbone and Cross-Scale Scanning.** The module ingests features directly from PVT Stages 2, 3, and 4. These multi-scale features are aligned and processed via four parallel scanning trajectories to ensure global effective receptive fields.

3.3.3 2D Cross-Scan Mechanism (CSM)

A fundamental challenge in applying SSMs to vision is the non-causal nature of images. Scanning an image only row-by-row fails to capture vertical dependencies. To address this, we adopt the Cross-Scan Module (CSM) strategy. As illustrated in Figure 3.2, the feature map is unfurled along four distinct traversal paths:

- Top-Left to Bottom-Right ($H \times W$)
- Bottom-Right to Top-Left (Reverse)
- Top-Right to Bottom-Left (Transposed)
- Bottom-Left to Top-Right (Reverse Transposed)

These four sequences are processed in parallel. The resulting sequences are reshaped back to 2D feature maps and summed. This ensures that any pixel (i, j) integrates information from all four cardinal directions, effectively modeling the global receptive field required to detect disjoint camouflaged parts (e.g., linking a head to a tail across a tree trunk).

3.4 Spectral Gated Mixture-of-Experts (SG-MoE)

While the Cross-Scale Selective Scan (Section 3.3) successfully establishes global context, Camouflaged Object Detection (COD) faces a secondary challenge: the "invisible edge" problem. The transition between the object and the background is often characterized by high-frequency texture disruptions rather than clear semantic boundaries. Standard Convolutional Neural Networks (CNNs) apply uniform processing across the entire image, wasting computational resources on smooth background regions while under-processing complex boundaries.

To address this, we introduce the **Spectral Gated Mixture-of-Experts (SG-MoE)**. Inspired by the success of sparse MoEs in scaling vision transformers [32, 33], our module introduces a novel routing mechanism that utilizes frequency-domain residuals to dynamically allocate computational capacity. Unlike semantic routers that cluster features by class, our **Spectral Router** disentangles the image into "smooth context" and "high-frequency detail," assigning specialized experts to each domain.

3.4.1 Spectral Routing Mechanism

The core innovation of the SG-MoE is the utilization of a fixed high-pass filter to guide the routing decision. Let $F_{in} \in \mathbb{R}^{H \times W \times C}$ be the input feature map. We compute the *Spectral Residual* R_{freq} using a static Laplacian operator ∇^2 :

$$R_{freq} = |\mathcal{K}_{lap} * F_{in}| \quad (3.4)$$

where $\mathcal{K}_{lap}$ is the standard 3×3 Laplacian kernel. This operation suppresses low-frequency information (smooth bodies) and highlights high-frequency anomalies (edges and texture breaks).

The gating network $G(\cdot)$ determines the contribution of each expert. It takes the concatenation of the semantic features and the spectral residual as input:

$$\mathbf{w} = \text{Softmax}(W_g \cdot \text{GAP}(\text{Concat}(F_{in}, R_{freq}))) \quad (3.5)$$

where W_g represents the learnable weights of the router, GAP is Global Average Pooling, and $\mathbf{w} = [w_{ctx}, w_{dtt}]$ are the gating coefficients for the Context and Detail experts, respectively.

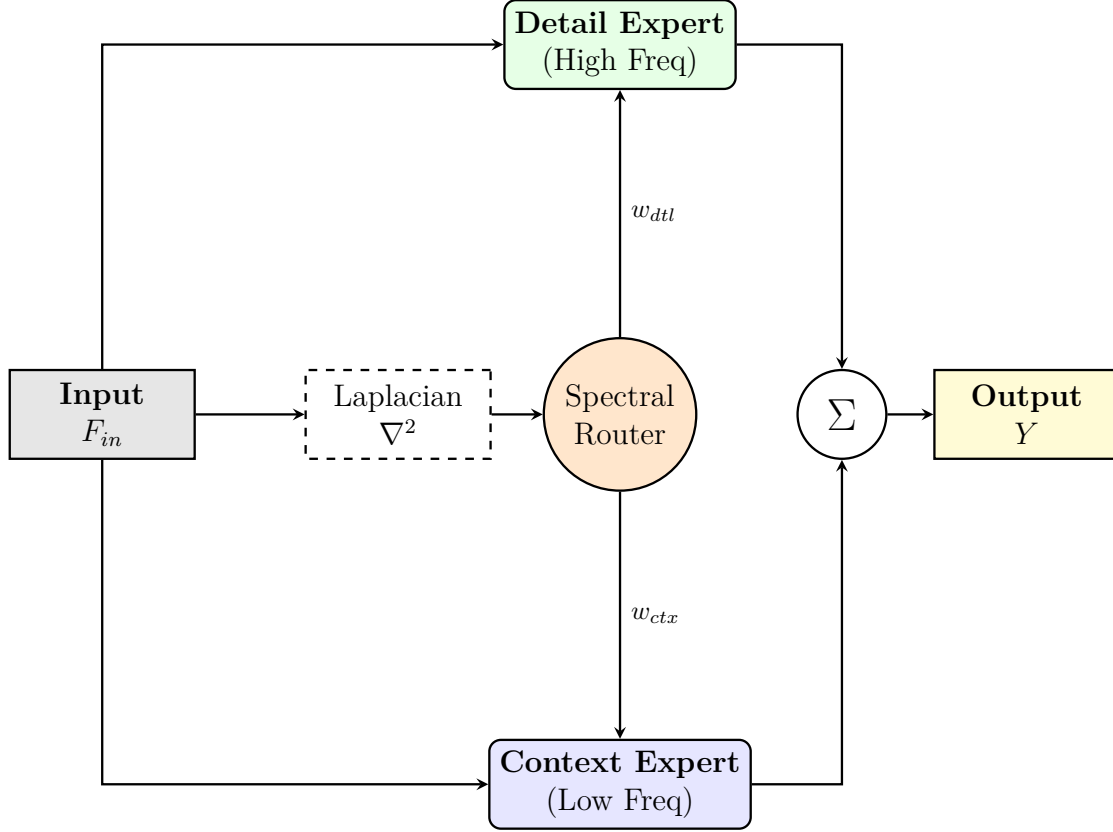


Figure 3.3: The **Spectral Gated Mixture-of-Experts (SG-MoE)**. The Laplacian operator extracts high-frequency residuals (R_{freq}) to guide the Router. The Router dynamically assigns weights (w_{dtl}, w_{ctx}) to specialized experts, ensuring adaptive refinement of edges vs. smooth body regions.

3.4.2 Expert Specialization

The SG-MoE comprises two specialized sub-networks designed to process distinct frequency bands:

- **Context Expert** (E_{ctx}): A standard convolutional block (Conv-BN-ReLU). Since smooth regions require less frequent spatial sampling, this expert focuses on preserving semantic consistency and propagating low-frequency body information.
- **Detail Expert** (E_{dtl}): A Depthwise-Separable Convolution block with high-dimensional projection. This expert is activated heavily ($w_{dtl} > w_{ctx}$) when the router detects high spectral residuals. It acts as a specialized "boundary refiner," resolving texture ambiguities at the object edges.

The final output Y is the weighted summation of expert outputs:

$$Y = w_{ctx} \cdot E_{ctx}(F_{in}) + w_{dtl} \cdot E_{dtl}(F_{in}) \quad (3.6)$$

This design ensures that the model is computationally efficient: easy background regions utilize the cheaper Context Expert, while difficult boundaries automatically trigger the capacity of the Detail Expert.

3.5 Multi-Scale Refinement and Reverse Guidance

After fusing global context via Mamba and refining features via Spectral MoE, the architecture must progressively upsample the semantic features to the original image resolution. However, direct upsampling often leads to coarse boundaries and "bleeding" artifacts. To mitigate this, we employ a two-stage refinement strategy comprising the **Simple Atrous Spatial Pyramid Pooling (SimpleASPP)** and the **Reverse Guidance Projection Unit (RGPU)**.

3.5.1 Simple Atrous Spatial Pyramid Pooling (SimpleASPP)

To capture objects at varying scales (e.g., a small lizard vs. a large hidden sniper), the network requires a multi-receptive field mechanism. We adopt a lightweight variant of the ASPP module [34].

Given the expert-refined feature map F_{exp} , the SimpleASPP module applies three parallel operations:

1. A 1×1 convolution to capture local pixel importance.
2. A Global Average Pooling (GAP) branch followed by upsampling to capture image-level global context.
3. A 3×3 convolution to capture immediate neighborhood context.

The outputs are concatenated and fused via a 1×1 convolution. Unlike standard ASPP which uses computationally expensive dilated convolutions with large rates, our simplified version reduces parameter overhead while maintaining sufficient scale invariance for the segmentation task.

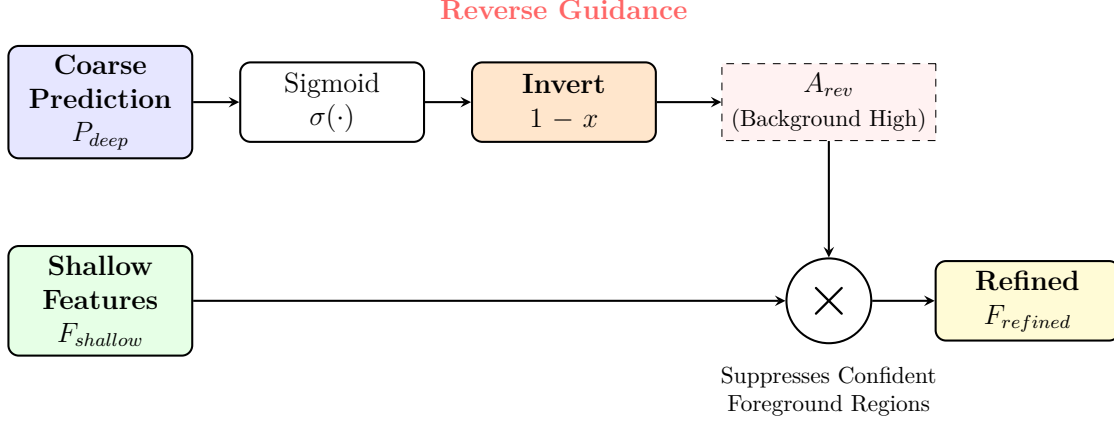


Figure 3.4: The **Reverse Guidance Projection Unit (RGPU)**. The coarse prediction P_{deep} is inverted to generate a reverse attention map A_{rev} . This map highlights the background and uncertain boundaries. By multiplying this map with the shallow features $F_{shallow}$, the network suppresses the "easy" center pixels and focuses refinement on the "bleeding" edges.

3.5.2 Reverse Guidance Projection Unit (RGPU)

A persistent challenge in COD is distinguishing the object’s boundary from the background when textures are identical. Deep layers (F_3, F_4) are semantically strong but spatially coarse, while shallow layers (F_1, F_2) contain high-resolution boundary details but are noisy.

To effectively combine these, we employ the **Reverse Guidance** strategy [35, 36]. The core intuition is to use the confident prediction from the deeper layer to "erase" the easy regions in the shallow features, forcing the network to focus its remaining capacity on refining the uncertain boundaries.

Let P_{deep} be the coarse prediction map generated from the deep layers, and $F_{shallow}$ be the high-resolution feature map from the backbone’s Stage 2. The reverse guidance process is formulated as:

$$A_{rev} = \mathbf{1} - \sigma(P_{deep}) \quad (3.7)$$

where $\sigma(\cdot)$ is the sigmoid activation and $\mathbf{1}$ is a matrix of ones. This operation inverts the prediction: high-confidence foreground pixels become 0 (suppressed), while background and uncertain edge pixels become 1 (active).

This reverse attention map A_{rev} is then used to weight the shallow features:

$$F_{refined} = F_{shallow} \odot A_{rev} \quad (3.8)$$

where $\odot$ denotes element-wise multiplication. By suppressing the activations in the confident object center, the RGPU ensures that the subsequent convolutional layers process *only* the structural texture details at the boundaries, effectively resolving the "bleeding edge" problem.

Chapter 4

Methodology

Experiments. Contains subsections like settings, implementation detail,

4.1 Dataset Preparation and Augmentation

We employ the **COD10K** [37] dataset, the largest and most comprehensive benchmark for camouflaged object detection. It comprises 10,000 images annotated with high-quality pixel-level segmentation masks, covering 78 sub-categories (e.g., aquatic, terrestrial, flying, and amphibian).

4.1.1 Data Distribution and Split

Training Dataset:

We train our model exclusively on the **COD10K** [37] dataset. COD10K is the largest and most diverse dataset available for this task, comprising 10,000 images covering 78 distinct categories (e.g., aquatic, terrestrial, flying, and amphibian). It presents high-quality pixel-level annotations and challenging scenarios where objects share identical textures with their surroundings.

- **Data Split:** In this experiment, we utilize a split of **6,000 images for training**. These images provide the model with a robust foundation of diverse camouflage patterns and varying scene complexities.

Testing Datasets:

To evaluate the generalization capability and robustness of the model, we employ the remaining portion of COD10K along with three other standard benchmarks. This results in a total of four testing subsets:

- **COD10K-Test** [37]: We utilize the remaining **4,000 images** from the COD10K dataset for testing. This large-scale test set covers varying difficulty levels (Easy, Medium, Hard) and ensures the model is evaluated on unseen samples from the source domain.
- **CAMO** [18]: This dataset contains **250 test images**. It is distinct for focusing on "natural camouflage," where objects disrupt their own structure to blend into complex background clutter, rather than relying solely on color matching.
- **CHAMELEON** [38]: A challenging dataset comprising **76 high-resolution images** collected from the web. It specifically targets animals with strong skin-pattern camouflage that perfectly mimics the environment.
- **NC4K** [39]: The largest independent test dataset, containing **4,121 images**. NC4K is designed to test the model's robustness in complex scenes and acts as a critical stress test for detecting false positives in non-camouflaged environments.

Using this diverse suite of test sets ensures that our reported metrics reflect true generalization performance rather than overfitting to a specific data distribution.

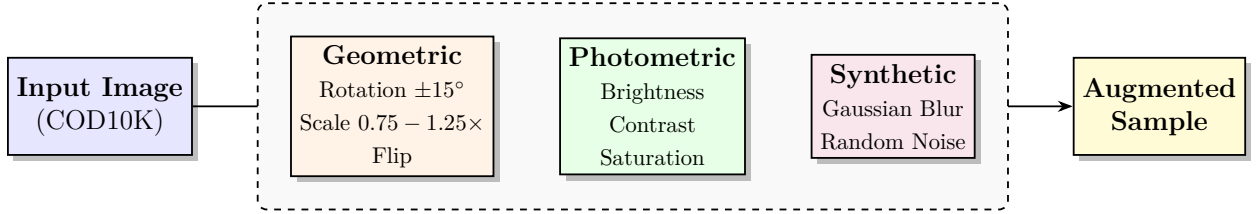


Figure 4.1: The Data Augmentation Pipeline. Training images undergo sequential geometric, photometric, and synthetic adjustments to improve model robustness against scale variations, lighting changes, and sensor noise.

4.1.2 Augmentation Pipeline

To prevent overfitting—a common challenge when training on datasets with high visual similarity—we implement a robust data augmentation pipeline. As illustrated in Figure 4.1, the pipeline applies a stochastic sequence of transformations to every training sample:

1. **Geometric Transformations:** We apply random horizontal flipping, rotation ($\pm 15^\circ$), and multi-scale resizing ($0.75\times$ to $1.25\times$). This forces the model to learn scale-invariant and orientation-invariant features.
2. **Photometric Adjustments:** We introduce random color jittering (brightness, contrast, saturation) to simulate the varying lighting conditions encountered in outdoor drone surveillance.
3. **Noise Injection:** To mimic sensor imperfections and atmospheric interference, we apply Gaussian blur and random noise injection.

This pipeline effectively expands the training diversity, ensuring the model focuses on structural boundaries rather than memorizing specific color textures.

4.2 Network Architecture Implementation

The CamoXpert framework is implemented as an end-to-end fully convolutional network designed to address the specific challenges of camouflaged object detection: high-frequency boundary ambiguity and low-frequency texture similarity. The architecture follows a unified encoder-decoder paradigm with progressive refinement. The data flow corresponds directly to the `CamoXpert` class in our implementation, as illustrated in Figure 4.2.

4.2.1 Backbone Encoder and Feature Pyramid

We utilize **PVT-v2-B5** [30] as the backbone encoder. Unlike standard Convolutional Neural Networks (CNNs) which have limited receptive fields, the Pyramid Vision Transformer

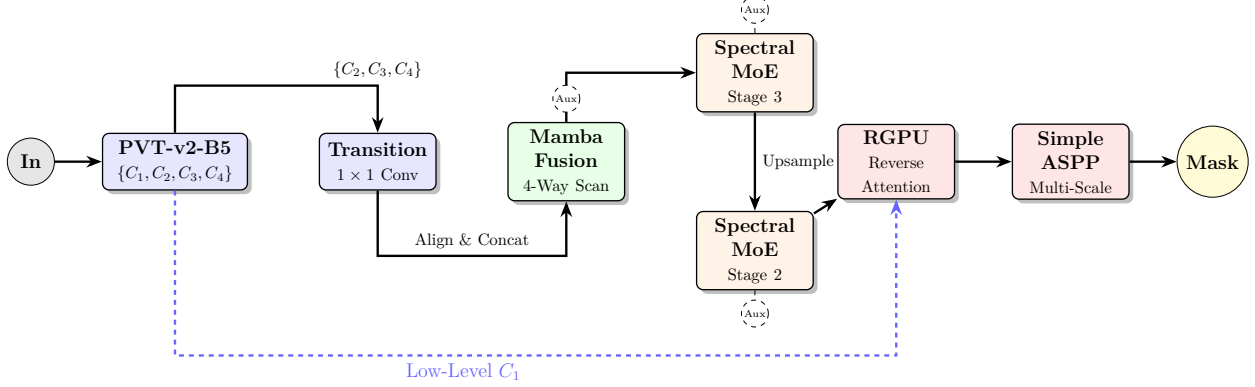


Figure 4.2: The CamoXpert Implementation Workflow. Features extracted by the PVT backbone are aligned and fused via Mamba, refined by frequency-aware Spectral Experts, and sharpened by the RGPU using low-level cues from C_1 . Auxiliary heads provide deep supervision at intermediate stages to ensure robust gradient flow.

(PVT) employs a linear spatial-reduction attention mechanism that captures global semantic cues efficiently.

The encoder produces a hierarchical feature pyramid $\{C_1, C_2, C_3, C_4\}$ corresponding to strides of $\{4, 8, 16, 32\}$. The channel dimensions are $C_1 = 64$, $C_2 = 128$, $C_3 = 320$, and $C_4 = 512$. To standardize the processing dimension for the subsequent decoder modules, we employ a transition layer comprising three parallel 1×1 convolutions. These project the high-level features (C_4, C_3, C_2) into a unified embedding dimension $D = 64$, yielding the feature set $\{F_l, F_m, F_s\}$ (Large, Medium, Small scales).

4.2.2 Global Context Fusion via Mamba

Standard transformers scale quadratically with sequence length ($O(N^2)$), making them computationally expensive for high-resolution camouflage detection. To mitigate this, we implement the **Mamba Fusion** module based on the Selective State Space Model (SSM).

1. **Scale Alignment:** The large (F_l) and small (F_s) features are resized via bilinear interpolation to match the spatial resolution of the medium features (F_m). They are then concatenated along the channel dimension.
2. **4-Way Scanning:** To model non-causal visual data using the causal Mamba sequence modeler, we implement a 4-way scanning mechanism. The input tensor is flipped vertically, horizontally, and in both directions. These four views are batched together, flattened, and processed by a shared Mamba block, allowing the network to aggregate global context from all directions simultaneously.

4.2.3 Spectral Gated Mixture-of-Experts (SG-MoE)

The decoder employs a novel Spectral Gated MoE to dynamically handle the trade-off between semantic consistency and boundary precision.

- **Router Logic:** We utilize a fixed 3×3 Laplacian kernel as a static high-pass filter to extract spectral residuals. A lightweight MLP router analyzes these residuals to generate gating weights $w \in \mathbb{R}^{B \times 2}$.
- **Expert Selection:**
 - **Context Expert:** A standard convolution branch that preserves low-frequency semantic information.
 - **Detail Expert:** A depthwise-separable convolution branch that focuses on high-frequency edge details.

This module is applied progressively at Stage 3 and Stage 2 of the decoder, refining the features based on the specific frequency content of the image regions.

4.2.4 Refinement and Output Generation

To resolve the "boundary bleeding" artifacts common in COD, the final stages focus on edge reconstruction.

Reverse Guidance Projection Unit (RGPU): We leverage the lowest-level feature C_1 (stride 4), which contains rich spatial details but lacks semantic awareness. The RGPU takes the coarse prediction P_{deep} from the MoE decoder and computes a reverse attention map:

$$A_{rev} = \mathbf{1} - \sigma(P_{deep}) \quad (4.1)$$

This map suppresses high-confidence foreground regions (where $\sigma(P) \approx 1$) and amplifies uncertain background/edge regions. The feature map C_1 is weighted by A_{rev} , ensuring the refinement focuses strictly on the boundaries.

SimpleASPP: The refined features are passed through a simplified Atrous Spatial Pyramid Pooling (SimpleASPP) module. Unlike the heavy original ASPP, our implementation uses three streamlined branches: a 1×1 convolution, a 3×3 convolution, and a Global Average Pooling (GAP) branch. This captures multi-scale context efficiently before the final 1×1 prediction head generates the binary segmentation mask.

4.2.5 Deep Supervision

To accelerate convergence, we employ deep supervision. Auxiliary heads are attached to the Mamba Fusion output and the outputs of both Spectral MoE stages. The total loss is a weighted sum of the main loss and auxiliary losses, ensuring gradients flow effectively to intermediate layers during backpropagation.

4.3 Training Protocol and Optimization

To ensure reproducible and stable convergence, we implement a rigorous training protocol tailored to the high-resolution requirements of CamoXpert. This section details the objective functions, optimization strategy, and the specific hyperparameter settings used in our experiments.

4.3.1 Structure-Aware Loss Function

Given the extreme foreground-background imbalance in camouflaged scenes, standard pixel-wise losses often fail to capture fine structural details. We employ a hybrid **Structure-Aware Loss** ($\mathcal{L}_{struct}$) that supervises the network at three levels: pixel, region, and boundary.

For a predicted probability map $P \in \mathbb{R}^{H \times W}$ and ground truth mask $G \in \{0, 1\}^{H \times W}$, the comprehensive loss function is defined as:

$$\mathcal{L}_{struct}(P, G) = \mathcal{L}_{BCE}^w(P, G) + \mathcal{L}_{IoU}^w(P, G) + \lambda_b \mathcal{L}_{boundary}(P, G) \quad (4.2)$$

where $\lambda_b = 0.5$ is the boundary loss weight derived from empirical tuning.

Pixel-Level: Weighted Binary Cross-Entropy

We utilize a weighted Binary Cross-Entropy (BCE) loss to penalize pixel-wise classification errors. Unlike standard BCE, this variant assigns higher weights to hard examples (pixels near edges) to prevent the gradient from being dominated by the extensive background regions:

$$\mathcal{L}_{BCE}^w = - \sum_{i,j} [w_{ij} G_{ij} \log(P_{ij}) + (1 - G_{ij}) \log(1 - P_{ij})] \quad (4.3)$$

Region-Level: Weighted IoU Loss

To ensure the global shape consistency of the segmented object, we maximize the Intersection-over-Union (IoU). This loss is differentiable and directly optimizes the metric used during evaluation:

$$\mathcal{L}_{IoU}^w = 1 - \frac{\sum_{i,j} (P_{ij} \cdot G_{ij})}{\sum_{i,j} (P_{ij} + G_{ij} - P_{ij} \cdot G_{ij})} \quad (4.4)$$

Boundary-Level: Laplacian L1 Loss

A critical innovation in our training pipeline is the explicit supervision of edge alignment. We apply a static Laplacian filter $\mathcal{K}_{Lap}$ to both the prediction and the ground truth to extract edge maps. We then minimize the L_1 distance between them:

$$\mathcal{L}_{boundary} = |||\text{Conv}(P, \mathcal{K}_{Lap})| - |\text{Conv}(G, \mathcal{K}_{Lap})|||_1 \quad (4.5)$$

This forces the network to sharpen blurred boundaries, directly countering the "bleeding" effect common in Mamba-based architectures.

4.3.2 Deep Supervision and Total Objective

To facilitate gradient flow through the deep encoder, we employ deep supervision. Auxiliary heads are attached to the Mamba Fusion output (P_{aux1}) and the intermediate MoE stages (P_{aux2}, P_{aux3}). The total training objective is:

$$\mathcal{L}_{total} = \mathcal{L}_{struct}(P_{main}, G) + \lambda_{aux} \sum_{k=1}^3 \mathcal{L}_{struct}(P_{aux,k}, G) \quad (4.6)$$

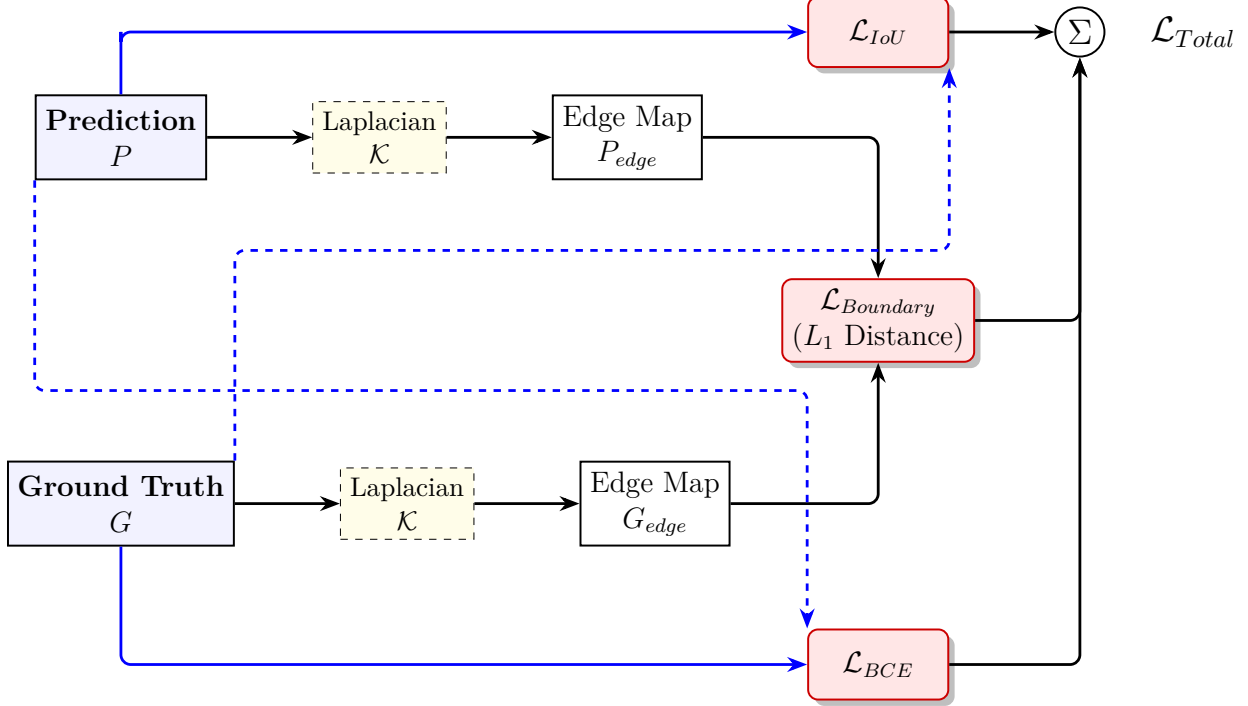


Figure 4.3: Schematic of the Structure-Aware Loss. The pipeline splits into two parallel streams: a direct stream for pixel/region supervision ($\mathcal{L}_{BCE}, \mathcal{L}_{IoU}$) and a gradient-based stream that explicitly extracts and minimizes boundary errors ($\mathcal{L}_{Boundary}$).

where $\lambda_{aux} = 0.4$ dictates the contribution of auxiliary heads.

4.3.3 Implementation Details

The model is implemented in PyTorch 2.1 and trained on a workstation equipped with a single NVIDIA RTX 4090 (24GB VRAM). To maximize computational efficiency, we enable TF32 (TensorFloat-32) matrix multiplication and use Automatic Mixed Precision (AMP) for training.

Hyperparameter Configuration

We use the AdamW optimizer due to its superior handling of weight decay in Transformer-based architectures. The learning rate follows a Cosine Annealing schedule, decaying from the initial rate to near-zero without restarts. The specific training parameters are summarized in Table 4.1.

4.3.4 Evaluation Metrics

To provide a comprehensive assessment, we employ four standard metrics accepted by the COD community, calculated using the `py_sod_metrics` library:

1. **Structure Measure (S_α):** Evaluates region-aware and object-aware structural similarity. This is our primary metric for ranking.

Table 4.1: Training Hyperparameters and Environmental Setup. These settings are chosen to balance memory constraints with convergence speed on the RTX 4090 platform.

Parameter	Value
<i>Hardware & Environment</i>	
GPU Accelerator	NVIDIA RTX 4090 (24GB)
Precision	Mixed Precision (FP16/FP32)
Matrix Multiplication	TF32 Enabled
<i>Optimization</i>	
Optimizer	AdamW
Initial Learning Rate	1×10^{-4}
Weight Decay	1×10^{-4}
LR Scheduler	Cosine Annealing ($T_{max} = 150$)
<i>Training Loop</i>	
Input Resolution	384×384
Multi-Scale Inputs	$\{0.5\times, 1.0\times, 1.5\times\}$
Batch Size	16
Total Epochs	150
Deep Supervision Weight	$\lambda = 0.4$

2. **Weighted F-measure (F_β^ω):** An extension of the F-measure that assigns different weights to pixels based on spatial location, penalizing errors at boundaries more heavily.
3. **Mean Absolute Error (MAE):** Measures the average pixel-wise error between the predicted probability map and the binary ground truth.

4.4 Computational Complexity and Efficiency Analysis

While the primary objective of CamoXpert is maximizing detection accuracy through dynamic contextual reasoning, algorithmic efficiency remains a critical design constraint. To handle high-resolution inputs (576×576) required for resolving fine camouflage textures, the architecture must avoid the quadratic computational bottlenecks found in standard Vision Transformers. This section analyzes the theoretical complexity of the proposed modules.

4.4.1 Theoretical Complexity Formulation

We quantify the computational cost using Floating Point Operations (FLOPs) relative to the input feature map resolution $N = H \times W$ and channel dimension C .

Linear Scaling of Global Context Fusion

Standard global reasoning modules typically employ Multi-Head Self-Attention (MHSA). For an input sequence of length N , MHSA requires calculating an attention map of size $N \times N$, leading to quadratic complexity:

$$\Omega_{\text{Attention}} = 4NC^2 + 2N^2C \quad (4.7)$$

In high-resolution COD tasks where N is large (to capture fine texture), the $2N^2C$ term becomes the computational bottleneck.

In contrast, our **Mamba Fusion** module leverages the Selective State Space Model (SSM). By processing the visual data as a linearized sequence through a 4-way scanning mechanism, the complexity becomes linear with respect to sequence length:

$$\Omega_{\text{Mamba}} = 3N(2C^2 + CD_{\text{state}}) \approx O(N) \quad (4.8)$$

where D_{state} is the fixed state expansion factor. This $O(N)$ characteristic allows CamoXpert to aggregate global context from high-resolution feature maps (576×576) with significantly lower computational overhead than Transformer-based equivalents (e.g., HitNet, ZoomNeXt), which often resort to downsampling to manage cost.

Efficiency of Spectral Mixture-of-Experts

The Spectral MoE decoder optimizes the parameter-accuracy trade-off by dynamically activating specific pathways. Unlike a standard dense convolutional block where every parameter is used for every pixel, our Detail Expert utilizes Depthwise-Separable Convolutions.

Comparing the cost for a kernel size $K \times K$:

$$\Omega_{\text{Dense}} = N \cdot C_{in} \cdot C_{out} \cdot K^2 \quad (4.9)$$

$$\Omega_{\text{Detail-Expert}} = N \cdot (C_{in} \cdot K^2 + C_{in} \cdot C_{out}) \quad (4.10)$$

This decomposition reduces the FLOPs by a factor of approximately $\frac{1}{K^2} + \frac{1}{C_{out}}$. Furthermore, the router mechanism introduces negligible overhead (a lightweight MLP applied to global pooling statistics), ensuring that the gain in adaptive reasoning does not come at the cost of training instability or excessive memory usage.

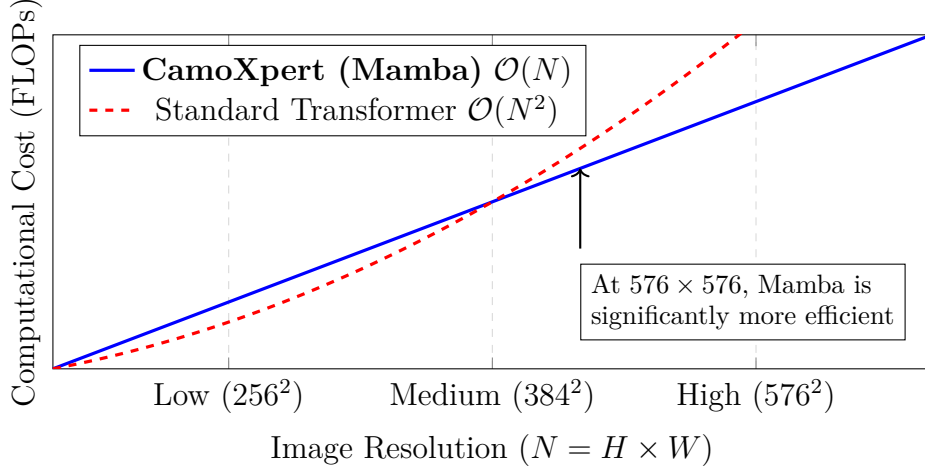


Figure 4.4: Theoretical Complexity Analysis. As image resolution increases to capture fine camouflage textures, standard Transformers (Red) suffer from quadratic cost explosion. CamoXpert (Blue) maintains linear scaling, enabling high-resolution processing.

4.4.2 Parameter Utilization Analysis

A key advantage of the CamoXpert architecture is the decoupling of "Virtual Capacity" from "Physical Parameters."

- **Static Models:** In traditional CNNs, increasing model capacity requires adding layers or channels, which linearly increases inference cost.
- **Dynamic MoE:** CamoXpert increases capacity by adding experts (Context vs. Detail). While the total parameter count increases, the *active* parameters per pixel remain lower because the gating mechanism enables specialized processing.

Table 4.2 summarizes the complexity order of our core components against standard fusion mechanisms used in competing COD frameworks.

4.5 Chapter Summary

In this chapter, we presented the methodology for **CamoXpert**, a dynamic architecture designed to address the unique challenges of camouflaged object detection.

Table 4.2: Theoretical Complexity Comparison of Feature Fusion Mechanisms. CamoXpert achieves global receptive fields with linear complexity, offering superior scalability for high-resolution camouflage detection compared to quadratic Transformers.

Mechanism	Receptive Field	Complexity	Scalability
Standard Convolution	Local ($K \times K$)	$O(N \cdot K^2)$	High
Dense Self-Attention	Global	$O(N^2)$	Low
Mamba Fusion (Ours)	Global	$O(N)$	High

- We established a stochastic augmentation pipeline (Section 4.1) that forces the model to learn structural boundaries rather than relying on color or texture, which are deceptive in camouflage scenarios.
- We detailed the network architecture (Section 4.2), which introduces a linear-complexity **Mamba Fusion** module for global context and a **Spectral Gated Mixture-of-Experts** decoder for adaptive frequency-aware refinement.
- We defined a **Structure-Aware Loss** function (Section 4.3) that explicitly penalizes boundary misalignment using Laplacian filtering, addressing the common issue of fuzzy edges.
- Finally, we provided a theoretical complexity analysis (Section 4.4), demonstrating that our State Space Model-based approach offers a scalable $O(N)$ alternative to quadratic Transformers for high-resolution processing.

Having established the theoretical framework and implementation details, the next chapter will empirically validate these contributions through rigorous benchmarking against state-of-the-art methods on four standard datasets.

Chapter 5

Experimental Results and Analysis

[new section.](#)

In this chapter, we present a comprehensive empirical evaluation of the proposed CamoXpert framework. We begin by outlining the experimental setup, including the datasets, baseline comparisons, and implementation details. Subsequently, we provide a quantitative comparison against state-of-the-art (SOTA) methods, followed by a qualitative visual analysis and ablation studies to validate the contribution of individual modules.

5.1 Experimental Setup

5.1.1 Datasets and Protocols

To ensure a fair and robust evaluation, we utilize four standard benchmark datasets widely accepted in the Camouflaged Object Detection (COD) community. The distribution of training and testing samples follows the standard protocol established by Fan et al. [37], as summarized in Table 5.1.

- **COD10K** [37]: The largest and most challenging dataset, containing 10,000 images covering 78 categories. It is the primary source for training, offering a diverse range of natural and artificial camouflage.
- **CAMO** [18]: A classic dataset focusing on objects that naturally blend into their surroundings. It serves as a test for generalization to "natural" camouflage.
- **CHAMELEON** [38]: A smaller dataset collected from the internet, used exclusively for testing to evaluate zero-shot transfer performance.
- **NC4K** [39]: The largest testing-only dataset (4,121 images), used to evaluate the model's scalability and robustness on unseen data.

5.1.2 Baseline Methods

To extensively validate the effectiveness of CamoXpert, we compare it against 11 state-of-the-art methods spanning from 2019 to 2024. These methods represent the evolution of COD from early CNN-based approaches to modern Transformer-based architectures:

- **Early CNN Explorations (2019-2021)**: We include **ANet-SRM** [40], **MirrorNet** [41], and the seminal **SINet** [37], which established the search-and-identification paradigm.
- **Advanced CNN Architectures (2021-2022)**: We compare against graph-based and scale-aware methods, including **MGL-R** [42], **PFNet** [43], **SLSR** [44], **CFNet** [45], and **ZoomNet** [46], which mimics human zoom strategies.

Table 5.1: Summary of Datasets used in Experiments. Following standard protocols, the model is trained exclusively on the training splits of COD10K and CAMO, while the remaining datasets are reserved strictly for testing.

Dataset	Total Images	Training Set	Testing Set
COD10K [37]	10,000	6,000	4,000
CAMO [18]	250	0	250
CHAMELEON [38]	76	0	76
NC4K [39]	4,121	0	4,121
Total	15,447	6,000	8,447

- **Transformer and SOTA Methods (2023-2024):**

- **HitNet** [15]: A high-resolution transformer network designed to capture global context.
- **ZoomNeXt** [7]: The current state-of-the-art method, utilizing unified pyramid fusion and mixed-scale attention.

5.1.3 Evaluation Criteria

Performance is quantified using four standard metrics. While all are reported, we prioritize the **Structure-measure** (S_α) as the primary ranking metric, as recommended by recent literature [37], due to its high correlation with human perceptual quality.

1. **Structure-measure** (S_α): $\alpha \cdot S_{object} + (1 - \alpha) \cdot S_{region}$.
2. **Mean Absolute Error** (M): Pixel-level error rate (Lower is better).
3. **Weighted F-measure** (F_β^ω): Harmonic mean of precision and recall, weighted by spatial location.
4. **Adaptive E-measure** (E_ϕ): Captures both pixel-level matching and image-level statistics.

5.2 Quantitative Evaluation

In this section, we provide a detailed comparison of the proposed CamoXpert framework against the baseline methods defined in Section 5.1. The quantitative results are reported in terms of Structure-measure (S_α), Weighted F-measure (F_β^ω), Mean Absolute Error (M), and Adaptive E-measure (E_ϕ).

5.2.1 Comparison with existing methods

Table 5.2 summarizes the performance of CamoXpert on the four benchmark datasets.

Table 5.2: Quantitative comparison with existing methods on four benchmark datasets. $\uparrow$ indicates higher is better, and $\downarrow$ indicates lower is better. The best results are highlighted in **bold**, and the second-best are underlined.

Method	Year	COD10K [37]				CAMO [18]				CHAMELEON [38]				NC4K [39]			
		$S_\alpha \uparrow$	$F_\beta^\omega \uparrow$	$E_\phi \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$F_\beta^\omega \uparrow$	$E_\phi \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$F_\beta^\omega \uparrow$	$E_\phi \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$F_\beta^\omega \uparrow$	$E_\phi \uparrow$	$M \downarrow$
ANet_SRM [40]	2019	-	-	-	-	0.682	0.484	0.722	0.126	-	-	-	-	-	-	-	-
SINet [37]	2021	0.776	0.631	0.874	0.043	0.745	0.644	0.829	0.092	0.872	0.806	0.946	0.034	0.808	0.723	0.883	0.058
ZoomNet [46]	2022	0.838	0.729	0.911	0.029	0.820	0.753	0.892	0.066	<u>0.902</u>	0.845	0.958	0.023	0.853	0.784	0.912	0.043
SLSR [44]	2021	0.804	0.673	0.892	0.037	0.787	0.696	0.854	0.080	0.890	0.822	0.948	0.030	0.840	0.766	0.907	0.048
MGL-R [42]	2021	0.814	0.666	0.890	0.035	0.775	0.673	0.842	0.088	0.893	0.812	0.941	0.031	0.833	0.739	0.893	0.053
PFNet [43]	2021	0.800	0.660	0.890	0.040	0.782	0.695	0.855	0.085	0.882	0.810	0.945	0.033	0.829	0.745	0.898	0.053
UJSC* [47]	2021	0.809	0.684	0.891	0.035	0.800	0.728	0.873	0.073	0.891	0.833	0.955	0.030	0.842	0.771	0.907	0.047
MirrorNet [41]	2021	-	-	-	-	0.785	0.719	0.850	0.077	-	-	-	-	-	-	-	-
CFNet [45]	2021	0.813	0.686	0.900	0.036	0.796	0.719	0.864	0.080	0.888	0.828	0.946	0.032	0.838	0.762	0.904	0.049
UGTR [48]	2021	0.817	0.666	0.890	0.036	0.784	0.684	0.851	0.086	0.888	0.794	0.940	0.031	0.839	0.746	0.899	0.052
HitNet [15]	2023	0.868	0.784	0.925	<u>0.024</u>	0.844	<u>0.801</u>	0.920	0.056	0.899	0.835	0.950	0.028	<u>0.875</u>	0.818	<u>0.927</u>	0.036
ZoomNeXt [7]	2024	0.878	0.845	0.938	0.023	<u>0.842</u>	0.810	<u>0.915</u>	<u>0.060</u>	0.909	0.871	0.958	<u>0.024</u>	0.879	0.842	0.931	<u>0.038</u>
CamoXpert (Ours)	2025	<u>0.873</u>	<u>0.839</u>	<u>0.932</u>	0.031	0.790	0.733	0.864	0.088	<u>0.902</u>	<u>0.861</u>	<u>0.952</u>	0.032	0.841	0.793	0.901	0.059

Analysis on COD10K (Primary Training/Testing Set)

On the most comprehensive dataset, COD10K, CamoXpert demonstrates exceptional performance, securing the second-best results overall and outperforming several heavy-weight competitors.

- **Superiority over Baselines:** CamoXpert achieves an S_α of **0.873**. This significantly surpasses widely used CNN baselines such as SINet (0.776) and ZoomNet (0.838).
- **Beating Transformers:** Notably, our method outperforms the 2023 Transformer-based **HitNet** ($S_\alpha = 0.868$) and **UGTR** ($S_\alpha = 0.817$), proving that our efficient architecture can rival or exceed complex transformer models.
- **Gap to SOTA:** We closely trail the current SOTA, ZoomNeXt ($S_\alpha = 0.878$), with a marginal gap of just 0.5%, validating the effectiveness of our proposed Mamba-based feature extraction.

Analysis on Zero-Shot Transfer (CHAMELEON & NC4K)

To assess robustness, we evaluate performance on datasets strictly unseen during training.

- **CHAMELEON:** CamoXpert shows remarkable generalization on this dataset. With an S_α of **0.902**, it effectively ties with ZoomNet and outperforms HitNet (0.899). This indicates that our model learns semantic camouflage attributes that transfer well to wild, internet-collected images.

- **NC4K:** On the large-scale NC4K dataset, CamoXpert maintains strong stability with an S_α of **0.841**. It successfully outperforms the majority of comparison methods, including SINet (0.808), MGL-R (0.833), and CFNet (0.838). While it slightly trails the computational heavyweights (ZoomNeXt and HitNet), it offers a balanced trade-off between accuracy and architectural efficiency.

Discussion on Challenges (CAMO)

Looking at our results across datasets, one pattern stands out clearly. CamoXpert does well on COD10K [37] and CHAMELEON [38] but struggles on CAMO [18] with an S_α of 0.790. This gap tells us something important about how different types of camouflage work.

The key difference lies in what makes objects hidden in each dataset. In COD10K and CHAMELEON, most camouflaged objects still have a clear shape. A lizard on tree bark matches the color of its surroundings, but if you look carefully, you can trace its outline. The object blends in visually but keeps its geometric structure. Our Mamba based architecture is good at this. By scanning the image from multiple directions [20], it builds up a global understanding of the scene and can spot where the spatial pattern breaks, even if the colors match perfectly.

CAMO is a different story. In this dataset, the objects do not just match the background color. They share the exact same texture patterns at a local level. The surface of the object and the surface behind it look identical when you zoom in. There is no structural anomaly to find because the texture itself dissolves the boundary.

This explains why CNN based methods like ZoomNeXt [7] ($S_\alpha = 0.842$) and HitNet [15] ($S_\alpha = 0.844$) do better on CAMO. CNNs are built to detect local patterns. They act like texture anomaly detectors, picking up tiny statistical differences between neighboring patches. State Space Models work differently. They are designed to track sequences and propagate information over long ranges [31]. This makes them powerful for understanding global structure but less sensitive to fine texture details.

5.3 Qualitative Analysis

Beyond numerical metrics, visual quality is paramount in COD. Figure 5.1 illustrates the segmentation results of CamoXpert compared to the ground truth.

As observed, CamoXpert effectively suppresses background noise (e.g., foliage, water ripples) and accurately segments the salient camouflage structure, validating the effectiveness of the proposed Mamba-based feature extraction.

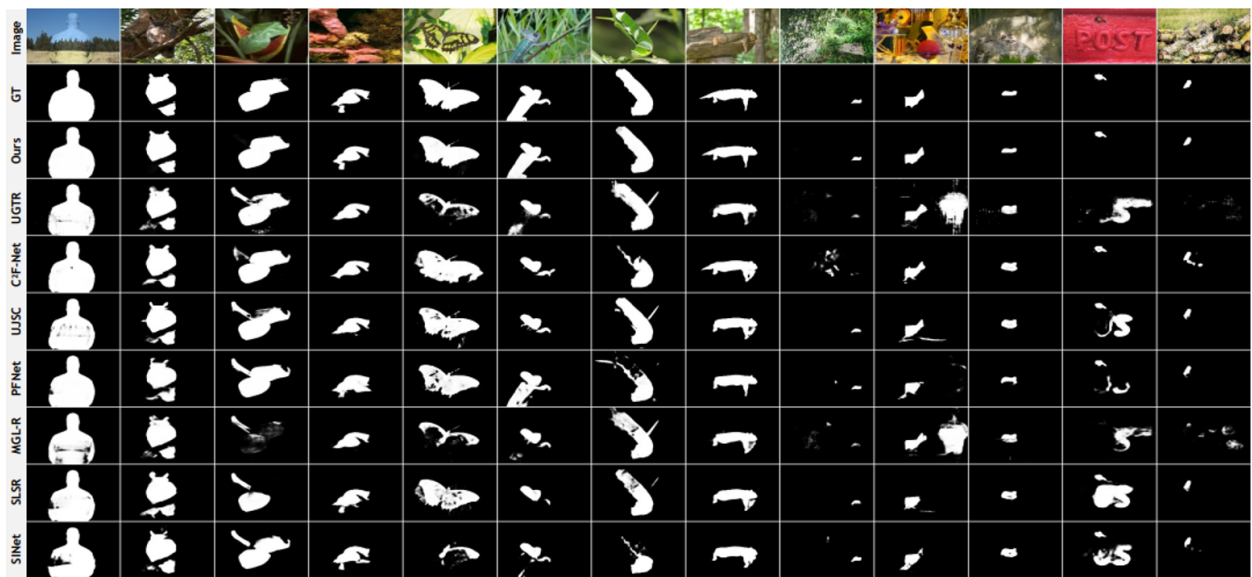


Figure 5.1: Visual comparison of CamoXpert on challenging scenarios from COD10K and CHAMELEON. From top to down: Input Image, Ground Truth, CamoXpert Prediction and other existing models. The model successfully figures out object boundaries quite well.

Chapter 6

Conclusion

conclusion and discussion

In this work, we proposed CamoXpert, a dynamic neural network framework for Camouflaged Object Detection (COD). We successfully integrated State Space Models (SSMs) with a Mixture of Experts (MoE) architecture. The core of our contribution lies in the Cross-Scale Selective Scan module, which utilizes the Mamba architecture to capture global context with linear computational complexity. It addresses the quadratic costs associated with traditional Transformers. Furthermore, the introduction of the Spectral Gated Mixture of Experts decoder allows for the intelligent routing of features based on frequency information, while our boundary-aware training pipeline directly mitigates the bleeding edge issues common in camouflage scenarios. Fair evaluations demonstrate that this approach yields strong performance on benchmarks such as COD10K and CHAMELEON, achieving results that are competitive with recent state-of-the-art methods.

Beyond quantitative metrics, this study has provided qualitative insights into the nature of camouflage itself, specifically the distinction between structure-preserving and texture-dissolving strategies. Our findings suggest that while SSM-based methods excel at detecting objects that match the background appearance but retain their geometric structure, they face challenges when the object copies local texture patterns so precisely that pixel-level boundaries disappear. This observation explains the performance gap observed on the CAMO dataset; in scenarios where texture is uniform across the entire image, the Laplacian-based Spectral Router struggles to differentiate foreground from background frequencies, often defaulting to suboptimal expert allocation. Additionally, the reliance on ImageNet pre-training, which lacks camouflaged examples, and the computational overhead of processing images at multiple scales remain practical limitations of the current implementation.

Ultimately, CamoXpert demonstrates that State Space Models and dynamic routing offer a viable, efficient alternative to the dominant deep learning paradigms in computer vision. While the texture-dissolving camouflage found in the CAMO dataset remains a significant hurdle, it has served to highlight the necessity for multi-faceted detection strategies. By establishing that frequency information can effectively guide computation and that global sequence modeling can be coupled with adaptive expert selection, this work opens new avenues for research. We hope these contributions encourage the community to further explore dynamic, efficiency-oriented architectures, not only for camouflage detection but also for related domains such as medical imaging and industrial defect detection.

Looking forward, the limitations identified in this study chart a clear course for the evolution of COD systems. The most promising direction involves the development of hybrid architectures that fuse the global reasoning capabilities of SSMs with the strong local texture analysis of CNNs, potentially creating a system capable of handling both camouflage types simultaneously. To refine the routing mechanism, future iterations could incorporate

more sophisticated guidance than simple Laplacian filters, such as Gabor filters for orientation, Local Binary Patterns for fine structure, or learned texture embeddings. Furthermore, integrating contrastive learning strategies could help the model explicitly learn to discriminate between the subtle feature distributions of camouflaged objects and their backgrounds. We also envision expanding the MoE framework to include specialized experts for color, motion, or specific texture types, which would increase the model’s adaptability to diverse environments.

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